From Chaos to Clarity: How AI Can Transform News into Actionable Intelligence for NFL Front Offices

Football
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1. Introduction

NFL teams specialize in speed, scale, and synergy. On the field, Next Gen Stats captures hundreds of data points per play [1], and teams manage billions of rows of structured data each season. Yet off the field, the daily flood of news—injuries, off-field behavior, and roster moves—still runs through outdated manual processes. During one-day waiver windows and action-packed pro-day workouts, infrastructure to efficiently process this information wins games.

Simultaneously, in the age of AI, teams are racing to enhance football operations without descending to ChatGPT [2] running the franchise. That means no to AI-written scouting reports and mysterious black-box projections—but yes to AI that amplifies the productivity of existing stakeholders by handling menial tasks that drain time and attention. Done right, AI is not about replacing football expertise but rather about complementing it effectively.

We unlock this advantage. Our research fills this role by translating unstructured *RotoWire* news reports [3] into decision-ready insights. We share a solution developed in partnership with a NFL team, the result of a multi-year collaboration in which we pioneered initial exploration, consulted with front office stakeholders, and built this novel system for responsibly integrating AI into football operations. We are proud to share that both buy-in and enacted feedback has not been limited to the analytics department but rather has come from a variety of areas in the organization.

The resulting architecture is a multi-stage framework that combines large language models (LLMs) for natural language understanding, Retrieval-Augmented Generation (RAG) [4] for contextual awareness across thousands of articles, and text-to-SQL capabilities to integrate unstructured text with structured metadata. A Chain-of-Thought (CoT) [5] reasoning layer decomposes complex queries into sequential steps, while vector embeddings enable weak-signal detection and preserve relationships beyond simple keyword matches. Furthermore, the entire system is locally hosted to guarantee the privacy and security of sensitive competitive intelligence.

Leveraging comprehensive news feeds, our system processes over 40,000 articles in minutes and can answer complex, multi-team, and context-rich questions. For our NFL partner, this system has the potential to automate custom reports and drastically reduce the time spent by scouts monitoring critical information, such as injuries and behavior, for nearly 2,000 draft prospects—a task that previously consumed multiple hours per day. We propose a foundational system and offer a series of validated use cases showing how AI can be responsibly used to streamline information curation and offer a competitive advantage.

1.1. Related Work and Motivation

Our research is primarily influenced by three areas of related work:

1. The latest state of artificial intelligence (AI), specifically the latest advancements in large language models (LLMs), the strong demonstrated performance of Retrieval-Augmented Generation (RAG), and the rise of highly interconnected agentic systems
2. The permeation of AI techniques into other flow sports in structured data capacities, such as advancements in soccer and fan-related analytics
3. A history of following football operations closely and past experiences doing research in more traditional areas in this space

While the third item is more of a mindset than something citable, we consider it important to discuss the first two areas and how they influence our approach.

1.1.1. Latest Research in LLMs, RAG, and Agentic Systems

The foundation of our work builds upon recent advances in large language models (LLMs), which have demonstrated remarkable capabilities in natural language understanding and generation. Modern LLMs such as GPT-4 [2], which exhibits human-level performance on various professional and academic benchmarks, and Claude 3 [6], which employs Constitutional AI for improved safety and alignment [7], have established new benchmarks for language understanding. Open-source models like Llama 3 [8], with its flagship 405B parameter model achieving comparable performance to GPT-4 on numerous tasks, and Qwen2.5 [9], which scales from 0.5B to 72B parameters and was trained on 18 trillion tokens, have democratized access to powerful language models. Most recently, DeepSeek-V3 [10] introduced a 671B parameter Mixture-of-Experts model that achieves state-of-the-art performance while requiring only 2.788M H800 GPU hours for training, demonstrating significant advances in training efficiency.

Retrieval-Augmented Generation (RAG) has emerged as a critical technique for enhancing LLMs with external knowledge, addressing their limitations in accessing up-to-date information and reducing hallucinations [4]. Recent advances in RAG have focused on improving retrieval quality and reasoning capabilities, with frameworks like Active RAG [11] determining when retrieval is necessary and hybrid RAG systems [12] that combine multiple retrieval strategies enhancing complex reasoning. Hierarchical approaches such as HIRO [13] employ recursive similarity scoring based on depth first search to minimize context delivered to LLMs without information loss, while EXIT [14] introduces extractive context compression that maintains contextual dependencies while enabling parallelizable extraction. These advances in RAG architectures directly inform our approach for processing and retrieving relevant news information from large corpora.

The integration of LLMs with structured data querying represents another crucial component of our system. Text-to-SQL capabilities have been significantly advanced through models like DAIL-SQL [15], which achieved 86.6% execution accuracy on the Spider benchmark [16], and DIN-SQL [17], which demonstrated that decomposing complex Structured Query Language (SQL) tasks into subproblems can improve LLM performance by approximately 10%. Recent work on agentic approaches, such as Agentar-Scale-SQL [18], implements orchestrated test-time scaling strategies that achieve 81.67% execution accuracy on the challenging BIRD benchmark [19]. These text-to-

SQL advances enable our system to seamlessly query structured metadata alongside unstructured news content.

The emergence of agentic LLM systems represents a paradigm shift in how language models interact with complex environments and execute multi-step tasks. Agent-based approaches have shown promise in various domains, with systems like ReWOO [20] introducing efficient reasoning workflows and SimRAG [21] demonstrating how retrieval can be augmented with iterative self-improvement. Chain-of-thought reasoning agents have proven particularly effective for decomposing ambiguous queries into sequential steps while maintaining context across retrieval operations [5]. Our system leverages these agentic capabilities to handle the complex, multi-faceted queries common in NFL front office operations and generate answers grounded in evidence to minimize hallucination.

1.1.2. AI and LLMs in Sports

The application of LLMs to sports analytics has gained significant traction, though most efforts have focused on consumer-facing applications rather than professional team operations. Recent work has highlighted the challenges LLMs face when processing sports narratives, with studies showing that even GPT-4o often fails to accurately aggregate basketball scores due to frequent scoring patterns and that open-source models like Llama-3 suffer from significant score hallucinations [21]. The SportsMetrics benchmark [22] introduced four novel tasks for evaluating LLMs' numerical reasoning and information fusion capabilities specifically in sports contexts, revealing fundamental limitations in how current models process sports data.

Comprehensive benchmarks have emerged to evaluate multimodal understanding in sports, with SPORTU assessing models across multi-level reasoning tasks from simple sports recognition to complex rule application. Even state-of-the-art models like Claude-3.5-Sonnet achieve only 52.6% accuracy on hard tasks requiring deep reasoning and rule-based understanding [23]. SCBench [24] specifically focuses on sports commentary generation for video content, while the MultiSports dataset [25] provides structured annotations for fine-grained action understanding across multiple sports.

Several efforts have explored sports-specific model architectures and training approaches. The GOAL benchmark [26] addresses knowledge-grounded video captioning for real-time soccer commentary generation, while K-SportsSum [27] combines knowledge graphs with summarization to generate sports news from live commentaries. Recent work on AI for handball [28] demonstrates the potential of combining deep learning with LLMs for predicting and explaining Olympic tournament outcomes.

The integration of multimodal capabilities has proven particularly challenging in sports contexts. SoccerAgent [29] implements a multi-agent system for comprehensive soccer understanding, achieving 85% accuracy on text-based questions but struggling with video-based reasoning at 60.9%. This performance gap highlights the difficulty of processing dynamic visual information alongside textual data. Work on universal soccer video understanding [30] has pushed toward more comprehensive frameworks, though significant challenges remain in real-time processing and multi-view analysis.

Despite these advances, there remains a notable gap in systems designed specifically for professional sports operations. Most existing work focuses on fan-facing applications like commentary generation or highlight creation, rather than the intelligence gathering and analysis needs of front office personnel. Our work addresses this gap by developing a system tailored to the specific requirements of NFL teams, focusing on actionable intelligence extraction rather than purely content generation. Unlike consumer-oriented applications that prioritize engagement, our system emphasizes accuracy, timeliness, and decision relevance—critical factors for competitive advantage in front office contexts.

The challenge of processing unstructured sports news at scale while maintaining factual accuracy has not been adequately addressed in prior work. Studies have shown that LLMs struggle with information aggregation in sports contexts, particularly when dealing with dense numerical data and temporal sequences [21]. Our approach of combining Retrieval-Augmented Generation (RAG) with text-to-SQL capabilities, validated against real front-office workflows, represents a novel solution to these challenges. Focusing on the specific information needs of NFL teams—injury monitoring, transaction tracking, and behavioral intelligence—we demonstrate that specialized systems can offer significant incremental performance utility in professional sports contexts.

2. Methods

In what follows, we discuss some technical details and design choice motivations in relation to each element of our final pipeline. While some readers may understandably find these details monotonous, we felt it was important to include them to validate the design of this system from a technical AI perspective and to aid in reproducibility and potential collaborative expansion.

2.1. Data

Central to building an agent to help front offices make sense of unstructured data is finding a suitable source of articles from which we can extract actionable insights. To make this possible, we partnered with *RotoWire* [3], the trusted leader in sports news and analysis, as the key information dataset that underlies our system. *RotoWire* has published millions of its articles since its founding nearly two decades ago, making it the perfect partner for building an AI system capable of extracting on-demand insights from unstructured data.

We view this partnership as somewhat analogous to partnering with the *Wall Street Journal* or *Financial Times* for a team building a finance-oriented agentic system; it's a match made in heaven in terms of having unstructured data ready ideal for mining insights. In addition to relying on *RotoWire* articles as its primary information source, our final proposed framework is designed such that it cites the specific *RotoWire* articles it used to arrive at its conclusions. We should be clear that, by unstructured data, we are not implying that *RotoWire*'s data is disorganized but rather using structure in the sense of its technical definition to explain that we are mining insights from prose instead of numeric or categorical columns in a relational database. To aid in reproducibility, we have written up detailed instructions in our README[1] on how to emulate our data collection process for any armchair analysts or research teams interested in building upon our system.

[1] See https://github.com/luh-j/football-news-AI.

The *RotoWire* data is ingested into our system through a customized extract, transform, and load (ETL) process. The purpose of this process is to continuously populate a PostgreSQL database with a PGVector-enabled schema that stores structured, time-stamped NFL news that supports vector embedding, reranking, and downstream LLM analysis. PGVector is a PostgreSQL extension that adds a vector data type, similarity search operators, and indexing methods, enabling the database to store embeddings and perform fast vector similarity queries for tasks like semantic search and RAG. The design of the ingestion system was guided by three central goals: reliability across long-date ranges, deterministic parsing of XML semi-structured data retrieved from the *RotoWire* API, and fast bulk inserts for database scalability.

The ingestion process begins by defining a controlled time-window schedule for *RotoWire* API requests for data retrieval. Because the *RotoWire* NFL News endpoint has practical limitations on how much historical data can be retrieved in a single request call, the system retrieves news in fixed 15-day chunks. To ensure long term reliability, this system is paired up with a rule that automatically triggers the ingestion function on 15-day intervals.

Once each 15-day window is retrieved, the pipeline performs deterministic XML parsing to transform the semi-structured *RotoWire* data into a normalized relational format. Every update entry is broken down into these core entries: timestamps, headline text, injury metadata, roster information, and associated player attributes. Through this structured schema of parsing data before inserting it into the PostgreSQL database, the system ensures that all records fed into the embedding and reranking subsystems follow uniform conventions, enabling stable vector representations and reliable document retrieval. The ingestion process then generates dense semantic vectors from the data using the MiniLM-L6-v2 sentence transformer model [31]. These vectors are then written back into the database, creating the vector-search layer required for retrieval, reranking, and agent-based analysis.

2.2. Initial Pipeline

In the context of a system now anchored in data, the key foundation of our system is the development of an agentic pipeline involving a collection of interconnected AI modules, on top of which sit several task-specific agents customized and tested for specific front office needs. The structure of this pipeline came into focus experimentation, initially starting with a simple sequential solution before escalating in an evidence-based manner to an interconnected, agentic workflow architecture. The initial sequential architecture is shown in Figure 1.

Our initial sequential framework consists of down-the-line progression through a reasoning stage followed by text-to-SQL, retrieval, reranking, and generation. At a high level, the reasoning stage interprets what is actually being asked, which is then converted at the text-to-SQL stage from intent into a series of executable database queries. The retriever processes the output returned by these queries, and the reranker orders these retrieved responses based on perceived relevance. The generator finally processes the highest-quality information highlighted by the reranker and forms the natural language prose reported back to the user.

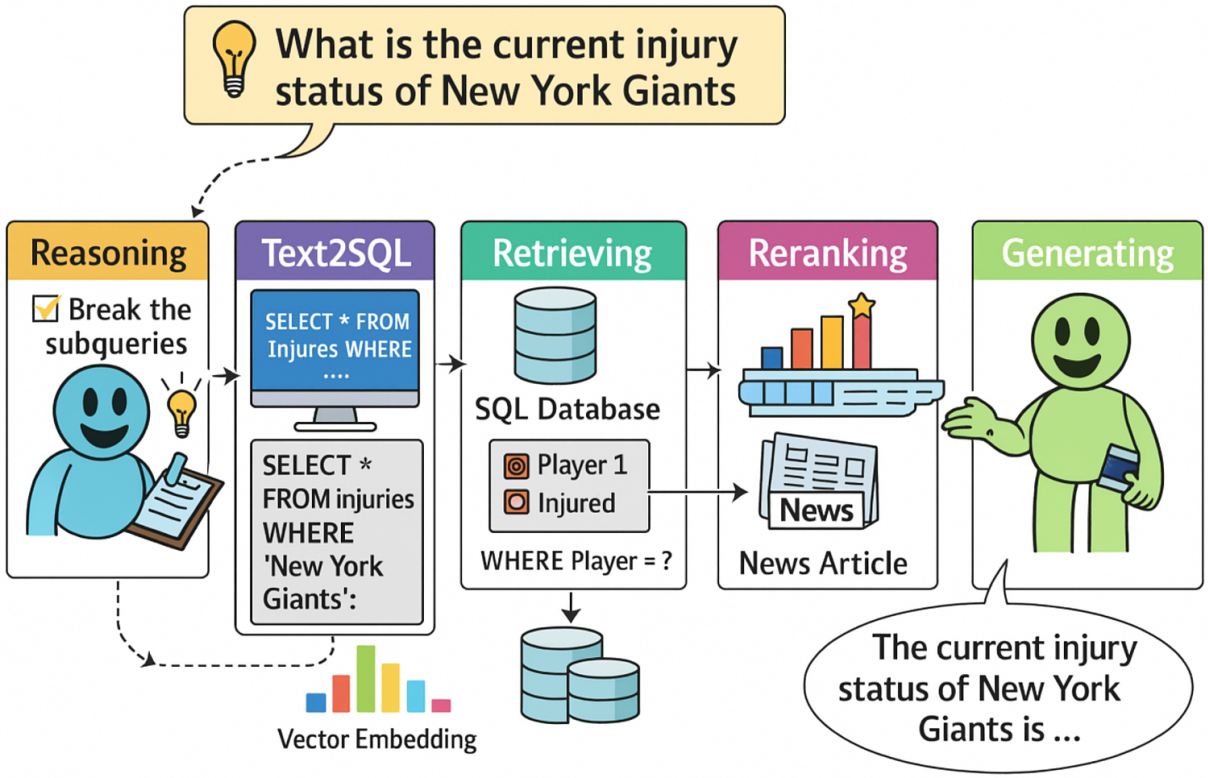


Figure 1: Initial Sequential Architecture*. A natural language query is processed through five sequential stages: (1)* ***Reasoning*** *decomposes the query into subqueries, (2)* ***Text2SQL*** *converts subqueries into database queries, (3)* ***Retrieving*** *fetches data from SQL databases and vector embeddings, (4)* ***Reranking*** *prioritizes relevant results from structured data and news articles, and (5)* ***Generating*** *produces a natural language response.*

While there are several stages in this process each focusing on different steps in the process, this setup is considered "sequential" because of the unidirectional, translational flow of information without any successive iterations. This was ideal as a starting point but proved to have some performance downsides in the context of generating verifiable, article-grounded content for front office needs. Several key issues including the following:

1. This system can generate informative responses, but its correctness is entirely at the mercy of finding the right *RotoWire* articles, ranking them appropriately, and not hallucinating any false information at generation time without safeguards or fact checks across stages. This is generally a prohibitively strong set of assumptions in the context of trying to build a tool to help front offices access information that we want them to be able to rely on as information for key decisions.
2. The sequential flow of information from stage-to-stage cuts off the ability of later components to revisit, improve, or correct earlier decisions, so errors and hallucinations—both in what gets retrieved and in how the final answer is written—can quietly propagate into the final output. By moving from this single-pass pipeline to an agentic workflow framework where reasoning, retrieval, generation, and verification agents share memory

and iteratively critique one another, we substantially reduce hallucinations at both the retrieval and generation levels and obtain more precise, higher-quality reports.

To demonstrate these issues more qualitatively, consider this example of output from the initial sequential pipeline when tasked with summarizing off-field behavior for a few players. The report looks believable enough on the surface but has a number of key issues under the hood:

Off-Field Behavior Report
Patrick Bryant: Patrick Bryant, Denver Broncos wide receiver, was involved in a minor altercation at a local restaurant on December 1, 2025, where he allegedly became belligerent and left without paying his bill, prompting an investigation by both the team and the NFL. **Malik Rivers** (College WR): Malik Rivers, a college wide receiver, was involved in a minor altercation with a store security guard at a local mall over a disputed parking spot on December 1st, 2025, but the University of Alabama's athletic department has not announced any disciplinary actions against him at this time.

***Example 1**: **Problematic Report from Initial Pipeline**. An off-field behavior report from our initial pipeline exhibits problematic hallucination. We use this as an initial baseline and, later in the paper, propose an improved pipeline that fixes the reflected issues.*

This report generally reads well, has good structure, and sounds reasonably convincing. However, it has errors that are classic issues of AI systems before additional safety steps are taken.

First, there's an issue with Patrick Bryant. While Bryant is a real player and did play for the Denver Broncos, there is no credible reporting that he was involved in the specific restaurant altercation described in the report. This entire scenario is made up on top of a real player's identity, which is a serious problem. In fact, we found that this agent would incorrectly tell you that just about anyone was in an altercation at a local restaurant or team gala and didn't pay the bill.

Second, Malik Rivers is not on the University of Alabama and is not a real player. There is a wide receiver named Malik Benson who was on the Alabama in 2023. It seems the initial system mixed up these two—likely due to lexicographical similarity—and hallucinated. This is a big problem.

Third, even if we give the system the benefit of the doubt and assume "Malik Rivers" was supposed to be a different real player with a lexicographically similar name, such as "Malik Benson," there is still the issue of improper incident identification. Similar to the Patrick Bryant case, the specific mall altercation described in the report is also not true regarding Malik Benson. This is still a significant performance issue that needs resolution if we want decision makers to be able to rely on this information.

Findings like this example confirmed that the simple approach of simply using Retrieval-Augmented Generation out of the box in a sequential pipeline likely wasn't going to cut it from a performance perspective. Accordingly, we iterated on these observations to design an agentic workflow system where we maintain the reliance on these reasoning, text-to-SQL, retrieval, reranking, and generation stages but turn them into more of a network than a sequence and have agents provide feedback on the output of other agents. Through this architecture, we achieve a

highly modular system with some self-correction and cross-checking capabilities that help with error reduction while also adding increased specialization that improves efficiency and quality. We will soon see that this yielded much more robust overall performance for the system.

2.3. Final Pipeline

Based on these updates, the final pipeline for the system now reflects an agentic workflow architecture with a verifier. This is less of a sequence and more of a network, and, to make it digestible, we visualize it as a number of specialized experts working together to provide a more nuanced answer to the question at hand.

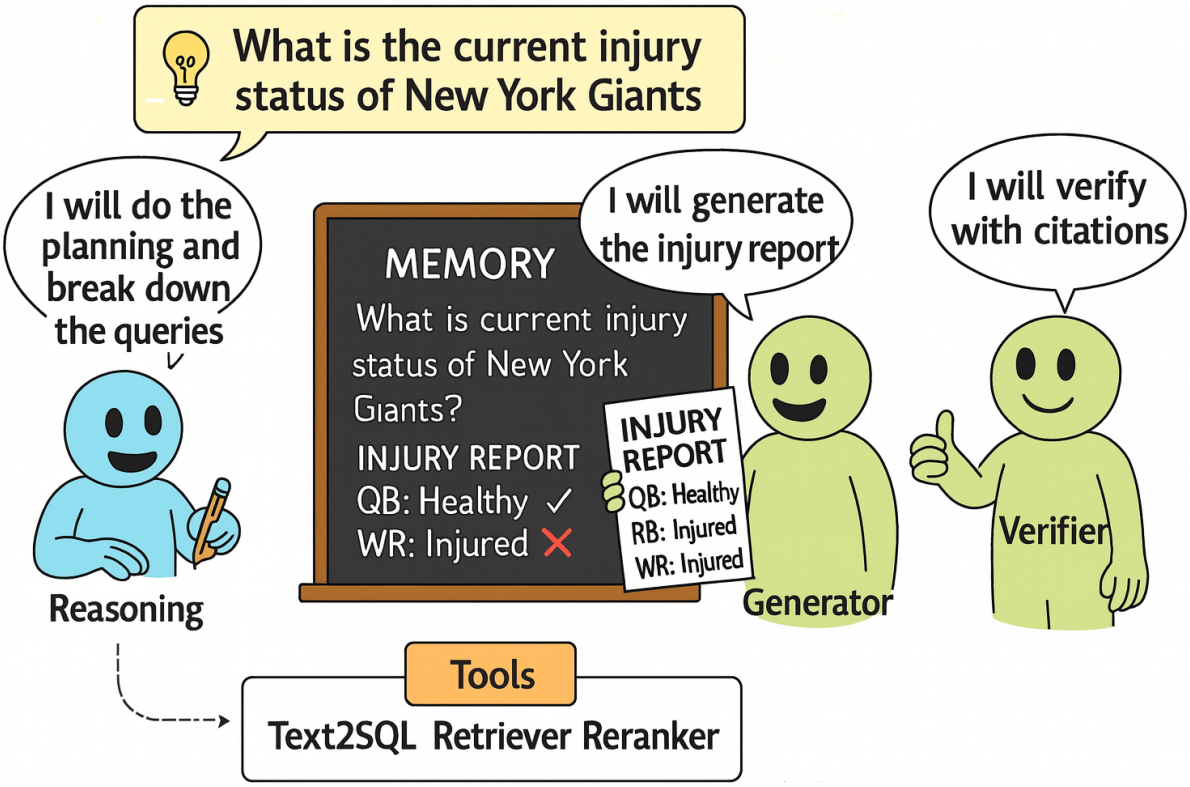


***Figure 2**: **Agentic Workflow Architecture—The Final Pipeline**. An agentic system with reasoning, generation, and verification stages coordinated through shared memory. The workflow enables iterative processing and dynamic tool invocation (Text2SQL, Retriever, Reranker), allowing the system to plan, execute, and self-verify responses with citations. This represents our final pipeline and constitutes a significant improvement on our flawed initial architecture from Figure 1.*

2.3.1. Verifier

A key addition to the final pipeline is the verifier, which is an internal quality-control agent of sorts that ensures every generated report remains factually grounded, internally consistent, citation-aligned, and free from any hallucinations. The verifier uses two separate agents that work cooperatively to safeguard against hallucinations, misaligned summaries, or unsupported claims.

After an initial draft report is prepared by an agent, it passes through its first checkpoint in the workflow in the form of the verifier agent. The verifier agent's role is to systematically examine each cited claim against the evidence that supposedly supports it. The process begins with a lightweight regex-based check over the draft, during which the agent scans for citation tokens and extracts the sentence immediately before each citation. This creates a mapping of each claim and the evidence it claims to reference. It also retrieves the evidence text from the structured metadata returned by the retriever corresponding to the citation. For each mapped sentence-citation text pair, the verifier makes an LLM call to perform a semantic analysis and judge the claims as either being (A) fully supported, (B) partially supported, or (C) unsupported. This feedback is stored and aggregated as a report and used to rework the report.

Once the verifier agent has finished evaluating and annotating a draft of a verification report, the editor agent takes over as the second stage of the workflow. The editor agent takes in three critical pieces of information:

1. Original retrieved context for preparing the initial report draft
2. The initial report draft produced by an agent
3. The verification report with information on each claim and whether the claim is supported.

This information is passed to another LLM, whose primary job is to reconcile these three items to create a more polished draft. The LLM's task is to rewrite the report to ensure that each and every claim is in strict compliance with the available evidence. All discrepancies identified by the verifier are removed if completely unsupported, and, in partially supported cases, the initial response is reworded to better reflect the ground truth. When claims in the initial draft are supported by the evidence, they remain in the report.

We consider the verifier to be important for several reasons. First, it cuts hallucination, which is the single biggest threat to our system, both in terms of performance in an AI sense and buy-in at an organizational level. Second, it not only ensures correctness but also offers source grounding via citation. This important both for attestation and empowering executives to efficiently access the information they might more directly be seeking. The system operates at the intersection of information curation and synthesization, offering flexibility depending on what is desired in the task at hand.

2.3.2. Retriever

In addition to adding the verifier, we took additional steps to increase the performance of each existing stage in the final pipeline by experimenting with additional design decisions for existing elements like the retriever. As for the technical design choices underlying the retriever, the embeddings are computed using all-MiniLM-L6-v2 [31][2], which is chosen for its strong semantic performance, low latency, and compact embedding dimensionality. The embeddings allow the system to scale to tens of thousands of documents without incurring heavy memory or inference overhead. Once the embeddings are computed, the system then carries out a K-nearest-neighbor (KNN) search over the embedding space, ranking stored documents by cosine distance to the query

[2] All-MiniLM-L6-v2 is a fine-tuned variant of MiniLM trained on over 1 billion sentence pairs using contrastive learning, developed during the Hugging Face Community Week. It is available at https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2.

embedding and returning the topmost semantically relevant matches. This gives us an initial set of candidates.

This final design is favored for its efficiency and semantic recall value. This search provides millisecond-level results even on larger datasets, which enables real-time retrieval for our agentic pipeline. Additionally, the search also captures contextual similarity, ensuring that reports effectively capture related injuries, player profiles, or medical updates even when surface wording differs substantially from the query.

To support downstream tasks, each retrieved record is parsed to create a structured metadata schema with information on player name, date, document identifier, a machine-generated citation identifier, notes, and analysis from the documents. This metadata of citations serves as the grounding for the generation of evidence-backed reports and ensures all claims are checked and tied down to a verified source.

In the process of building out the retrieval pipelines, several evaluation cycles were carried out to assess the precision and optimize the latency of the system. For the initial vector search, the most important metrics tested were recall and position among the k-most relevant documents. Recall measures how often relevant injury documents appear amongst the top-k returned results. Checking this ensures the retriever captures the full spread of semantically relevant contexts without missing any subtle reports. On the other hand, precision measures the proportion of top-k results that are actually relevant for report generation. This metric ensures that the system avoids surfacing generic sports content that could dilute context for reports created via agents. Our agents were tuned via experimentation to maximize performance on these fronts.

Beyond statistical performance metrics, validation was carried out by running real-world queries on the retriever system and manually cross-verifying retrieval quality. This hybrid evaluation process has helped ensure that the retriever consistently surfaced the most accurate and meaningful documents across player names and other unstructured phrasing variations.

2.3.3. Reranker

The retrieval system was built to address the precision and speed limitations of dense vector search. While the primary PGVector retrieval ensures high recall, completely relying on vector similarity often results in “hard negatives”—documents that have a high semantic overlap with the query text but don’t match its logical intent. These hard negatives are particularly problematic because they falsely answer the question at hand but often appear seemingly correct and similar to the query. To resolve this problem, the retrieval pipeline integrates a secondary semantic reranking stage.

The reranker design methodology favors cross-encoder architectures over other approaches, such as bi-encoders or late-interaction models. However, bi-encoders cannot capture subtle semantic nuances between the queries and the documents through token-level interactions, especially compared to late-interaction models. Since the retrieval process calls for capturing these semantic nuances, a different and better design is used. Through processing queries and documents simultaneously within a full attention mechanism, the cross-encoder models achieve far better semantic alignment compared to other architectures. Furthermore, the availability of these high-performing, efficient, and pretrained cross-encoders, which are compatible with libraries like

sentence-transformers, ultimately enabled a robust yet cost-effective implementation of the retrieval system. Compared to proprietary external LLM APIs that might also suffer from hallucinations and long generation times, it is also useful that most of these cross-encoders are open source. These open-source cross-encoders handle the reranker system.

The process for selecting our final reranker model involved tests designed to identify an architecture that is capable of parsing syntax generating semantic misalignment and relating to common NFL terminology. Standard embedding models often cannot distinguish between a "rumored trade" and a "completed transaction" or between the "projected" and "confirmed" injury status of a player, as the vector embeddings for these concepts are similar. To quantify the important ability to distinguish, a sample dataset with syntactically similar but logically different NFL news reports was used to test the performance of the candidate reranker models. The sample dataset features complex semantic traps, syntactic ambiguity, and conditional logic similar to the example above to ensure that the models are tested on whether they find the most relevant documents.

We evaluate reranker performance using two complementary metrics: top-1 accuracy and Normalized Discounted Cumulative Gain [32] at rank 5 (NDCG@5). Top-1 accuracy measures whether the most relevant document is ranked first, which is an important consideration for the system since the highest-ranked passages disproportionately influence the LLM's generated response [33]. Conversely, NDCG@5 evaluates the top-5 retrieved results, providing an overall assessment of ranking quality within this window while applying a logarithmic discount that penalizes relevant documents appearing in lower positions. NDCG@5 is particularly important since, under the condition of limited context windows, retrieving relevant documents but ranking them below irrelevant ones wastes valuable context space and can introduce noise that degrades generation quality. Together, these metrics offer both the precision needed to find the single best source and the overall ranking quality that determines what information the LLM sees when generating its response.

Quantitative analysis of six chosen reranker models revealed visible performance gaps between them. A baseline MiniLM model [31] achieved 0% for top-1 accuracy on five query cases and had a Normalized Discounted Cumulative Gain (NDCG@5) of 0.5909 using the sample NFL news dataset. The BGE-reranker-large model [34] had better results. It achieved 60% top-1 accuracy with an NDCG@5 score of 0.8000. The best candidate observed was the zerank-1-small reranker model [35], achieving 80% top-1 accuracy with an NDCG@5 score of 0.9262 for the sample dataset and demonstrating far better semantic precision compared to other models. This largely aligns with the overall general, non-football performance rankings of reranker models found online, with the zerank-1-small model being high up in the list.

Note that there exist better performing models than the zerank-1-small, such as its larger twin, the zerank-1 model [35]. However, we determined that the larger parameter size resulted in diminishing marginal returns given the insignificant performance gains as well as potential complications regarding deployment on AWS due to the large size of the models. We ended up choosing the zerank-1-small model due to its superior performance across overall football and non-football information and more manageable resource footprint.

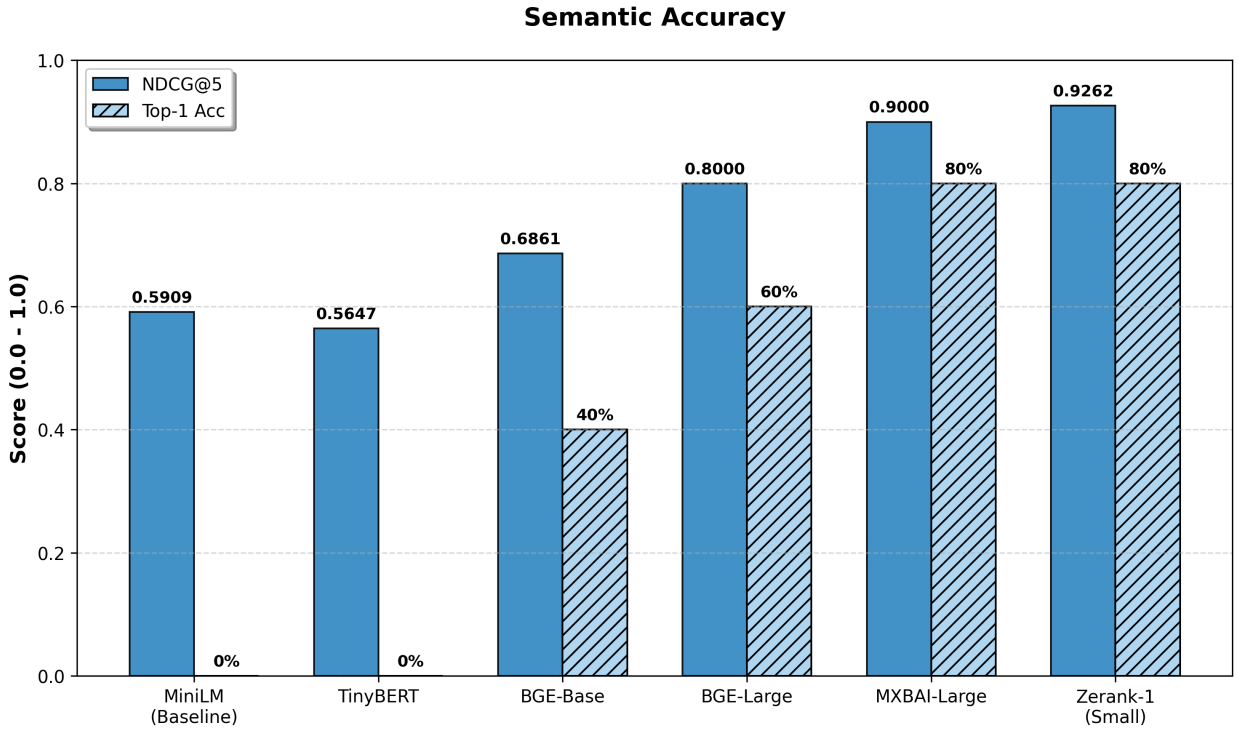


***Figure 3**: **Semantic Accuracy of Considered Reranker Options**. Comparative performance analysis of six reranker models evaluated on a sample dataset of five semantically complex NFL query scenarios. The metrics shown are Top-1 Accuracy (striped bars), which indicates the percentage of queries where the most relevant document was ranked first, and NDCG@5 (solid bars), which measures the overall quality of the ranking for the top five results. Each model was tested five times to achieve better illustration.*

The integration of a reranker model like the zerank-1-small model fundamentally improves the retrieval system's reliability in syntactically ambiguous situations—a performance gain which we consider to be particularly valuable for the system. This improved reliability was most evident in one test case from the sample dataset. We asked questions about specific playoff implications like "What happens to the Chiefs playoff seed if they lose to the Bills?" Lighter models incorrectly prioritized retrieved documents from cases where the Chiefs beat the Bills or documents that offered no information on playoff seeding implications. In other words, they often just highlighted documents that contained "Chiefs," "Bills", or "lose" without any regard for the directionality or whether playoff implications were discussed. For the zerank-1-small model, it was able to understand the conditional syntax and retrieved a document describing the outcome of the Chiefs losing the game to the Bills as was requested. In another case, we asked, "Who replaced Justin Reid after he was injured?" The lighter models had trouble understanding the subject, and they chose documents with Reid replacing other players instead of being the injured player or simply offered unrelated replacement updates for other players without mentioning Reid. For the zerank-1-small model, it successfully distinguished between the player being replaced versus the player doing the replacing. By incorporating an additional validation stage, the retrieval pipeline ensures that LLM generators and downstream agents can operate on factually correct and rigorous evidence.

2.3.4. Specialized Task-Specific Agents

To accomplish our research goal of enabling front offices to extract meaningful new information from unstructured data, we layer task-specific agents—namely injury, contracts, waiver, performance, transaction, and general off-field news agents—on top of this pipeline that encapsulates the recently discussed retriever and reranker (among several other stages). Each of these task-specific agents designed for particular use cases in a football operation.

Pragmatically, a significant step in building these agents was establishing interaction protocols among agents. Initial prototypes revealed that loosely defined inter-agent communication resulted in drift, redundancy, or inconsistent assumptions about task requirements. To address this, strict message formats and validation constraints were introduced. A coordinator agent was designed to be responsible for generating all sub-agent instructions using schema-based templates, ensuring that downstream agents always received unambiguous task definitions. Moreover, throughout the development process, behavioral constraints were applied to reduce hallucination and promote consistency. These constraints included explicit "rules of operation" embedded in system prompts, such as restricting claims to information present in the provided evidence, prohibiting speculative reasoning, and requiring uncertainty labeling when data were incomplete. The iterative refinement of these constraints proved essential: early versions of the injury agent (a task-specific agent used for tracking player health) tended to infer player statuses or injury severities without sufficient justification. After introducing stronger grounding requirements and clarifying the reasoning steps expected of the agent, these failures were substantially reduced.

Evaluation of the agent design involved repeated cycles of testing and revision. Synthetic test cases were created to probe edge scenarios, including contradictory injury reports, missing data, outdated information, and examples with players appearing in multiple contexts. In each cycle, errors were traced to either prompt structure, insufficiently explicit instructions, or overly permissive reasoning allowances. Adjustments were then applied to the agent specifications, leading to progressively more reliable behavior. Through this iterative process, the agents achieved stable performance across both well-structured and noisy inputs.

3. Results

The primary results of this paper take two forms:

1. First, a series of specialized task-specific agents that operate on top of the proposed agentic workflow pipeline to provide specific information in several football operation domains. We walk through a few demonstrative examples that highlight these agents' utility.
2. Second, focused motivation of the power of the underlying agentic workflow pipeline and how it positions us to develop organization-specific, customized solutions for specific information desired by decision makers.

3.1. Examples

Regarding the former, one task-specific agent developed and recently referenced is the injury agent which can keep tabs on player health. For this task, our development process focused on enabling the agent to interpret heterogeneous injury information while producing outputs with consistent

structure and terminology. Prompt engineering played a central role: multiple iterations were performed to optimize the agent's ability to (1) extract relevant injury events from unstructured text, (2) interpret medical terminology and severity indicators, and (3) standardize these findings into a predefined report schema. Templates were created to ensure uniform phrasing, explicit reasoning steps, and predictable JSON output. These prompts were refined through empirical testing, where failures, such as missed injuries, inconsistent formatting, or unsupported medical claims, were systematically analyzed and corrected.

To showcase the injury agent, we can explore what it would have said about Jayden Daniels after his elbow injury against the Seattle Seahawks in week 9 of the 2025 NFL season. The agent is able to automatically detect Daniels' injury and report that it is a dislocated elbow that will lead backup quarterback Marcus Mariota to start several games for the Washington Commanders.

Jayden Daniels Injury Analysis (December 5, 2025)
Report
Current Status: Jayden Daniels is currently listed as QUESTIONABLE with a left elbow injury [1]. He was medically cleared and practicing fully on October 1st, but the exact recovery timeline for his elbow injury is not specified. **Medical Analysis**: Daniels has sustained a left elbow injury, which is common in football, often due to repetitive strain or direct trauma. The typical recovery period for such an injury can range from 4 to 8 weeks, depending on the severity and the individual's healing capacity. The risk of re-injury is moderate, especially if the player returns too soon without full recovery. **Impact Assessment**: With Daniels QUESTIONABLE, the team's depth at quarterback is somewhat compromised. Marcus Mariota is set to start in Week 10, but Daniels' availability for the upcoming games is uncertain. This could impact the team's offensive strategy and performance. **Return Projection**: Based on the information provided, Daniels' return date is not specified. However, if he follows a typical recovery timeline, he might be ready by mid-December. The exact date will depend on the evaluation of his injury and the team's medical staff.

***Example 2**: **Jayden Daniels Injury Analysis**. The report provides an accurate overview of Daniels' injury situation, noting the nature of his injury, available recovery timeline information based on what was known at the time, and the injury's implications on the team.*

We can see that the report in general correctly characterizes the nature of Daniels' latest injury. It is clear on what his injury is and how it will impact his availability, and the report generally offers as much timeline information as was available at the time of generation. We see that the report gets a little touchy on referencing being medically cleared from a previous injury in the second sentence but still offers a sentence that is technically true and keeps the elbow injury as its main focus even if it links it to a previous injury recovery. Note that the "[1]" on the second line of the "current status" section is a citation the system has automatically generated to its top retrieved *RotoWire* article that emerges from the reranking process, specifically the article titled "Jayden Daniels: Coach confirms left elbow injury" published by *RotoWire* on November 2, 2025. This article is shown below in Example 3 and importantly reflects the ability to the system to not just answer questions but also performantly site where the content of the answer originates.

While the report here is itself exciting, it is also important the system can show its underlying process that led to its answer. Specifically, one of the best parts of this proposed final system is that we can see how the verifier considers claims and decides whether or not they are supported based on the retrieved documents. This process operates before the generation of the reported answer and effectively refines the answer to root out hallucination. Specifically, we can see the verifier confirms the notion established by the Retrieval-Augmented Generation phase of the pipeline that Daniels is indeed questionable because of a left elbow injury.

Jayden Daniels Injury Analysis (December 5, 2025)			
Claims			
Verifier Claim			**Verifier Verdict**
`Jayden Daniels is currently listed as questionable with a left elbow injury.`			`Supported`
Top-Three Retrieved Supporting Documents			
Document Rank	**Title**	**Date**	***RotoWire* Article Content**
1	Jayden Daniels: Coach confirms left elbow injury	11/2/25	*Head coach Dan Quinn confirmed that Daniels suffered a left elbow injury in Sunday's 38-14 loss to the Seahawks, but Quinn didn't have any further updates on Daniels' status, Nicki Jhabvala of The Athletic reports.* *Daniels needed a brace to stabilize his left, non-throwing arm after sustaining an injury in the fourth quarter. A more concrete recovery timeline for Daniels will likely be established following additional evaluation of the injury. Marcus Mariota is set to fill in as the starting QB beginning with Week 10 against the Lions.*
2	Jayden Daniels: Medically Cleared Practicing Fully	10/1/25	*Commanders coach Dan Quinn said Daniels (knee) has been medically cleared and will be a full participant in Wednesday's practice, Tashan Reed of The Washington Post reports.* *Daniels will return from his two-game absence due to a knee sprain and start Sunday's road matchup against the Chargers. It appeared as though the second-year signal-caller had at least a chance to return Week 4 versus Atlanta, so by this stage there should be minimal, if any, lingering concerns about Daniels' mobility and rushing upside.* *Meanwhile, Quinn also said things are "trending up" for wideouts Terry McLaurin (quadriceps) and Noah Brown (groin), per Nicki Jhabvala of The Washington Post.*
3	Jayden Daniels: Gets back to full Wednesday	10/29/25	*Daniels (hamstring) practiced fully Wednesday.* *Coach Dan Quinn told Tashan Reed of The Washington Post earlier Wednesday that Daniels was slated for a full week of practice, and he ended up kicking it off with no restrictions whatsoever in the first session of Week 9 prep. Daniels thus appears as if he'll miss just one game due to the low-grade right hamstring strain that he sustained Week 7 at Dallas. Friday's injury report will*

			unveil whether or not he enters the weekend with a designation ahead of Sunday's contest against the Seahawks.

Example 3: The Verifier's Role and Top Three Documents Shaping the Jayden Daniels Response*. We explore how different parts of the pipeline contribute to the overall response for Jayden Daniels. Our agent reports all of this information to the user in addition to the response shown in Example 2. Specifically, we see that the verifier decides that the central claim is that Daniels is questionable with a left elbow injury and concludes that this is a fully supported argument. We can see that the verifier picks an appropriate claim and arrives at the correct support conclusion. Separately, we also see the articles retrieved and order of importance assigned by the reranker (with only the top three articles shown for compactness). Note that the reranker is smart enough to understand that its second- and third-ranked articles do relate to Jayden Daniels injuries but don't address his active injury, which it correctly infers is the one that a front office likely cares about in relation to his upcoming availability. The reranker clearly situates the most topical article first in relation to Daniels' active injury. These steps enable the agent to offer strong explainability, and we see that the underlying processes accurately reflect the Daniels injury.*

Big picture, this is somewhat of an uninteresting finding on the surface. It's guaranteed that 100% of the 32 NFL teams were aware that Daniels had an elbow injury, and, while we don't know for sure as outside researchers, we strongly suspect that teams have strong automated internal infrastructure operating on league-provided data with up-to-date injury info in a relational database. Even if for some reason teams don't have this, we recognize that even the most cursory glance at *ESPN*, *The Athletic*, *X*, or any beat writer article would probably do the trick regarding Daniels specifically. After all, no one needs a complicated AI system to break the news that one of the most recognizable quarterbacks in the NFL is hurt.

However, we do still see a key use case of this service from conversations with our NFL partner that is underscored by this example. This circles back to a pivotal point about our research: we are not trying to use AI in a way that tells decisions makers things they already know or what to do. Instead, we are interested in arming executives with desired information in a way that is more accurate and efficient. Critically, this goal is as far as possible from the misguided idea of ChatGPT running a football team.

Specifically, we designed this injury service based on conversations with our partner noting how difficult this injury information can be to track for draft prospects (or, more generally, players who aren't in the NFL). Our partner estimated that they scout roughly 2,000 amateur players per year and that, during the peak of the amateur scouting season, keeping tabs on these prospects can take scouts north of two hours per day in the form of scouring local news publications and manually reading oodles of articles each week. Our partner emphasized that this is menial and time-consuming work that no one enjoys where a solution would be particularly impactful, which is exactly the specific type of problem we are trying to solve with this research. The idea is that the injury agent might not deliver much value in telling a team that Daniels is hurt but could deliver significant value in streamlining the process of monitoring the health of thousands of draft prospects.

Thinking in this state of mind of curating useful info and saving time where it matters, an additional agent we developed in consultation with our partner is a specialized service that expands on tracking player status and comments on team relations and off-field issues in addition to injury info.

Specifically, Example 4 (on the next page) showcases this agent's commentary on wide receiver Noah Brown of the Washington Commanders. In Example 4, the cited sources are the real-life *RotoWire*[3] articles that our system has retrieved that are used to drive its response. They are ranked from most important to least important as determined by the reranker. We have provided links to these articles so you can reference them and see how they guide the provided response.

It's worth noting that the ability to both ground the response in cited sources and understand that the article about Noah Brown's return to practice should guide the response more heavily than those about him previously being on injured reserve or logging limited practices is a strong testament to why using AI in a robustly curated way is appropriate for this task. The additional safety measures happening within the agentic workflow pipeline are central to this. Overall, the impactful value add here is that our pipeline makes it possible to not only answer this question but also to do so in a way beholden to ground truth articles and in a way that captures the big picture. We find it particularly notable that the agent is able to simply say that there are no in-scope documents supporting the notion that there are any problematic disciplinary, legal, or organizational risk factors, an area where many AI systems would be inclined to hallucinate like our initial pipeline did in Example 1.

Arguably, in terms of the value of this agent's commentary on NFL players specifically, we see more value in this report in an NFL context than the one on Daniels' injury. This report is conveying information that is almost certainly not readily able to be queried from a team's internal database in entirety, and, while this information could be obtained from reading articles in classic sports publications, there's value here in multiple sources of information and storylines that have clearly evolved over time being appropriately synthesized and referenced by our agent autonomously.

To elaborate on this in more detail, we think it is particularly valuable that this agent can detect and report on roster cuts and responsibly hypothesize on how off-field issues, health, or organizational dynamics could be meaningful context underlying such a decision. While the individual example here regarding Noah Brown is interesting and could be pertinent if a team was doing due diligence in preparing for free agency or a potential trade, there is larger importance here from a research perspective than is reflected in just one player. Noah Brown at the time of writing is in his second season with the Commanders and doesn't have any key transactional roster info included in his present-day response because he has been on the active roster without interruption for nearly two years[4]. However, when Noah Brown was cut two years ago by the Houston Texans after the 2024 preseason, this agent would have highlighted that he was cut and offered context about this move. We consider the ability to quickly provide this information for recently cut players to be a key area of functionality for this agent.

[3] Note that the articles contained at the links which involve *CBS News* are still written by *RotoWire* and mention this at the top of the *CBS News* pages.
[4] Brown did sign a new contract in the offseason, but this is not a key point here because he stayed with the Commanders, and there was little public fanfare surrounding the negotiation.

Noah Brown: Off-Field Overview

Report

1. **Factual Explanation**: Noah Brown, a wide receiver for the Washington Commanders, has been dealing with groin and knee injuries throughout the 2025 season. The team has listed him as limited or ruled him out for several games due to these injuries.

2. **Potential Impact**:

- **Availability**: Due to his ongoing injury issues, it's uncertain when Brown will be fully healthy and available for games.
- **Team Relations**: There is no indication of any tension between Brown and the team regarding his injury status or treatment.
- **Performance**: The frequency and severity of his injuries may impact his performance on the field.

3. **Disciplinary, Legal, or Organizational Risk Factors**: No information is available to suggest that Brown's injuries are related to disciplinary, legal, or organizational issues.

4. **Fantasy and Real-World Football Forecast**:

- **Fantasy**: Due to the uncertainty surrounding Brown's availability, his fantasy value may be affected. If he returns from injury, his performance will likely impact his fantasy relevance.
- **Real-World Football**: The Commanders' depth chart at wide receiver will need to be reassessed if Brown is unable to contribute due to his injuries. This could lead to increased playing time for other receivers on the team.

Cited Articles

Rank	Article Date	Article Title	*RotoWire* Article Link
1	10/15/25	Lands on IR	https://www.cbssports.com/fantasy/football/news/commanders-noah-brown-lands-on-ir/
2	10/11/25	Ruled out for Week 6	https://www.cbssports.com/fantasy/football/news/commanders-noah-brown-ruled-out-for-week-6/
3	10/2/25	Out for Week 5	https://www.cbssports.com/fantasy/football/news/commanders-noah-brown-out-for-week-5/
4	10/10/25	Listed as limited	https://www.cbssports.com/fantasy/football/news/commanders-noah-brown-listed-as-limited/
5	10/8/25	Back at practice	https://www.cbssports.com/fantasy/football/news/commanders-noah-brown-back-at-practice/

***Example 4*: *Noah Brown Off-Field Overview*.** *We see that the agent accurately comments on considerations that a team might care about in a transactional context for Noah Brown. The agent accurately characterizes Noah Brown's injury status and notes that there are no known active off-field issues surrounding Noah Brown. Importantly, we see the agent identifies when it doesn't have data on a field of interest rather than hallucinating an imaginary response.*

Throughout the preseason, teams are constantly cutting players, with each team moving from typically on the order of 90 total players at the start to 53 players after the last preseason game (with cuts heavily concentrated right after this last preseason game). As teams aggressively move towards finalizing their 53-man roster before the regular season, each team cuts tons of players, and there is a mad dash to figure out who is available and why… something we have discussed in detail with our NFL partner. Our understanding is that it is not uncommon for teams to convene all of their scouts and analysts for an all-nighter on this day since speed can be important for ensuring any diamonds in the rough who got cut land with your team rather than a competitor.

We designed this agent with cut day specifically in mind. This agent can autonomously and efficiently curate information on who has been cut and synthesize information on why immediately once articles on these transactions exist. While we are hesitant to belabor a point harped on frequently, the core idea here remains not having AI decide something like what players to add but rather letting AI curate information for executives in a way that saves time to create a competitive advantage. We have interest in developing agents specifically designed for organizational needs on key days on the NFL calendar and see this example as strong evidence that we have the infrastructure necessary to do so.

Big picture, and in relation to our second key result, the beauty of our overall framework is that these top-level, specialized agents are modularized and easily constructible. Specifically, we can refine existing agents or design additional agents to accomplish specific tasks desired by decision makers. One central point we are trying to convey here is that none of these specialized capabilities are the be all and end all… while we designed them to have legitimate utility to a football operation based on close coordination with our NFL partner and do believe that they have legitimate use cases, our key research finding is that we have designed a system that foundationally has the ability to tackle these types of use cases. For example, we are confident that we could design custom, organizational-specific services that operate on top of our agentic workflow pipeline to address specific information requests from a scout, general manager, analyst, strategist, or coach. We aim to use this paper as a launch point for developing additional services desired by teams (or even non-NFL entities, like writers, college teams, or player agencies) and are publishing this because we believe in the fundamental underlying structure and its ability to empower these goals.

3.2. Ablation Studies

So far, our discussion of results has been largely football-focused, an evaluation strategy we consider important given that the goal is to deliver information that could be qualitatively useful to members of a football operation. However, we did want to additionally evaluate some of our design decisions at the pipeline level as well to ensure our final chosen architecture was simultaneously responsible from an AI system design perspective.

To determine the performance gains yielded by our retrieval architecture, we conducted an ablation study comparing the final pipeline against a baseline standard vector search process using PGVectors only. The quality of the retrieval process is mainly defined by minimizing hallucination to ensure that the downstream agent does not generate incorrect reporting based on factually incorrect or outdated retrieved context.

Revisiting another earlier area of focus, we did ablation testing that specifically focused on the NFL's 53-man roster cutdown day. This event presents a distinct retrieval challenge characterized

by transactional ambiguity and potential for "hard negatives," namely when documents describing opposing actions share high semantic overlap, such as "waived" and "signed". We aimed to minimize agent hallucinations stemming from the retrieval of semantically unrelated or contradictory transactions. We evaluated the system's ability to find information on players waived to meet the 53-man limit using the following query: "Which players did the Chiefs waive or release to reach the 53-man limit in 2025?"
For the baseline scenario using PGVectors only without our latest design decisions, the system could not capture the transactional intent of the request. It only prioritized high-similarity general player news with high keyword density for "Chiefs" and generic roster status but failed to attend to the important verbs of "waive" and "release". The baseline scenario without our latest design decisions ranked documents describing player status updates, contract extensions, and draft picks as the most relevant. These include documents talking about things like the following:

- "Chiefs head coach Andy Reid said Thursday that Pacheco 'looks tremendous' at OTAs and has 'put on some good weight.'"
- "Kelce and the Chiefs agreed to terms Monday on a two-year extension that makes him the highest-paid tight end in the NFL."
- "The Chiefs selected Simmons (knee) in the first round of the 2025 NFL Draft, 32nd overall."

While this information is technically related to the roster, it provides little to no evidence regarding cut day departures. Most of the information is not even from around the time of roster cutdown day in question. The agent relying on this context would most likely report that no players were cut or even falsely list healthy players as being released if not handled carefully.

In contrast, the final pipeline successfully identified the correct documents explicitly describing the roster exits. The reranker in the final pipeline provides an additional level of semantic checking, filtering out signings and injury reports to show only documents with semantic information that are also present in waivers. The reranker successfully picks out documents that directly talk about players being waived or let go from the team, with three out of the top five documents reranked matching the timing of this year's cut day (August 26, 2025). These include documents describing the following:

- "Steele's opportunity with the Chiefs ends as the process of cutting down to a 53-man roster begins."
- "Smith was waived by the Chiefs on Tuesday. Smith signed with Kansas City on Aug. 1 after being let by go by the Raiders in late July, and reeled in one pass for eight yards, plus two kickoff returns for 62 yards over three preseason games. The wide receiver will likely clear waivers, and it's possible he joins the Chiefs' practice squad ahead of the 2025 season."
- "Oladokun was waived by the Chiefs on Tuesday. Oladokun signed a reserve/future contract with Kansas City in February after spending the majority of the 2024 campaign on the team's practice squad. The quarterback completed 11 of 23 passes for 120 yards and turned two carries into 36 rushing yards over three preseason contests with the team. If Oladokun clears waivers, he'll likely stick around as a member of the Chiefs' practice squad in 2025."

The final pipeline correctly attends to phrases like "opportunity with the Chiefs ends" and "waived by the Chiefs", aligning ideally with the request. These qualitative results confirm that the final pipeline can fundamentally change the logical validity of the retrieved context. By ensuring that the

agent receives precise evidence regarding which players are leaving the team rather than generic team news, we can prevent generating reports that incorrectly list cut players as active ones.

4. Conclusion

As 32 NFL teams race to capitalize on AI, we offer a measured solution for responsibly using AI to extract information that can help a football operation. Our research introduces a reproducible pipeline combining language understanding and context-aware retrieval with validation against real front-office workflows. We propose specific workflows for injury detection, player transactions, off-field happenings, and general performance. We believe these agents are useful in their own right but more importantly reflect the development of a foundational pipeline that can malleability answer a wide range of questions in this domain. From the thousand-foot view, we view this work as a part of the broader opportunity for AI to influence flow sports and, particularly, how it can play an effective role in leveraging free-form data with implications in under-integrated off-field contexts. Based on our consistent conversations with our NFL partner and seeing the insights that can be extracted from *RotoWire* articles, we believe that systems like this are the start of a movement towards responsible AI integration in football and beyond.

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